HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **low-level device drivers** for an advanced SoC platform, focusing on **memory management** and **processor subsystem interaction**.
- Developed and integrated **embedded software** within **RTOS environments** (Zephyr), managing real-time task scheduling and inter-process communication.
- Engineered **hardware-facing software capabilities**, prototyping and documenting functionalities for integration with cross-functional engineering teams.
- Enhanced **debugging and validation methodologies** for embedded systems, reducing the time to identify root causes of hardware/software interaction issues by approximately **30%**.
- Collaborated with **hardware architects** and firmware developers to troubleshoot and resolve complex issues related to **SoC performance** and **system integration**.
- Contributed to the development of **Python scripts** for automating testing and validation of embedded software components, improving test coverage efficiency.
- Containerized embedded software components using **Docker** for consistent development and testing environments on **AWS EC2** instances.

SKILLS

Core Languages – C, C++, Python, Assembly

RTOS & Embedded – Zephyr RTOS, FreeRTOS, ThreadX, Embedded Systems Architecture, Device Drivers, RTOS Architecture, Hardware Abstraction

SoC & Architecture – SoC Development, Processor Architecture, ARM, RISC-V, Multicore Systems, Cache Coherency, Inter-processor Communication (IPC)

Tools & Debugging – JTAG, Trace32, OpenOCD, Segger J-Link, LLDB, QEMU, Jenkins, Git, Github Actions, Gerritt

Operating Systems – Linux, Real-Time Operating Systems (RTOS)

Cloud & Containerization – AWS EC2, Docker

PROJECTS

Advanced RTOS Driver Development for Compute Platforms

- Developed critical **device drivers** for a multi-processor SoC using **C**, interfacing directly with hardware registers and memory maps.
- Implemented real-time task management and synchronization primitives within a **Zephyr RTOS** environment, ensuring deterministic execution for sensor data acquisition.
- Created comprehensive unit tests and integration test harnesses using **Python** to validate driver functionality and performance across various hardware configurations.

Embedded Software Automation and Integration Suite

- Built a suite of **Python** scripts to automate the build, deployment, and initial validation of embedded firmware on target hardware.
- Integrated automated testing workflows with **Jenkins** CI/CD pipelines, enabling faster feedback loops for embedded software development cycles.
- Designed and implemented a basic **inter-processor communication** (IPC) layer in **C** to facilitate data exchange between two distinct processor cores on an SoC prototype.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)